MILITARY SPECIFICATION

REPEATER, REGENERATIVE MX-9331( )/URC AND AUXILIARY EQUIPMENT

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for Repeater, Regenerative MX-9331( )/URC, and auxiliary equipments consisting of a vehicular power adapter, MX-9545( )/VR; and a battery case, CY-7499( )/U. The equipment is used to extend the range of tactical military radio sets for ground and airborne applications when operating in plain voice and with speech security equipment of the NESTOR and SAVILLE families.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

T-T-871	Twine, Cotton, Wrapping
NN-P-71	Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, N.J. 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-5810

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SPECIFICATION (cont'd.)

PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrguated, Triple Wall
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-S-901	Shock Test, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-M-38510	Microcircuits, General Specification for
MIL-V-55341	Valve, Pressure Relief
MIL-R-49107	Repeater, Regenerative MX-9331 ( )/URC, Nuclear Survivability Requirements for (U)

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STANDARDS

Federal

FED-STD-356	Commercial Packaging of Supplies and Equipment

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1275	Electrical Circuit, 28-Volt DC Transient Characteristics for Military Vehicles

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DRAWINGS

Electronics Command

DL-SM-B-687700	MX-9331( )/URC
DL-SM-B-687701	MX-9545( )/VR
DL-SM-B-687709	CY-7499( )/U
SM-C- 687871	TRANSFORMER, XMTR

(Copies of specifications, standards and drawings, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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environmental conditions for reliability testing. The following environmental conditions 4.7.2

3. REQUIREMENTS

3.1 Construction. The equipment shall be constructed in accordance with the requirements of this specification, and the Data List Drawings DL-SM-B-687700, Regenerative Repeater, MX-9331( )/URC: DL-SM-B-687701, Vehicular Power Adapter, MX-9545( )/VR; and DL-SM-B-687709, Battery Case, CY-7499( )/U, including the requirements for parts, materials and processes therein.

3.2 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.3 Parts, materials and processes, general. (See 4.4). The general requirements of MIL-E-5400 shall be complied with except that MIL-P-11268 shall be the prime specification for the selection of electronic parts, materials and processes, including microelectronics, instead of the paragraph entitled "Selection of Materials, Parts, and Processes and Associated Data" of MIL-E-5400. Should any electronic part, material or processes considered for usage, not conform with MIL-P-11268, it shall be submitted for approval per MIL-P-11268 procedures. All parts or materials (standard per MIL-P-11268 or non-standard) furnished by the contractor for airborne equipment shall meet the requirements of Category II, MIL-E-5400.

3.3.1 Microelectronic devices. Microelectronic devices shall be selected in accordance with MIL-M-38510, Class B.

3.4 Finish, protective. (See 4.4). The equipment shall be given a protective finish in accordance with MIL-F-14072 and the equipment drawings.

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3.5 Marking. (See 4.4).

3.5.1 General. Marking shall conform to MIL-M-13231 (see 4.4). Front panel marking shall be Group I as described in that specification.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Each Regenerative Repeater, MX-9331( )/URC, and auxiliary equipments shall be serial numbered.

3.6 Preconditioning.

3.6.1 Bounce preconditioning. (See 4.6.1). Each Regenerative Repeater, MX-9331( )/URC, and auxiliary equipments produced shall meet the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (also see 4.5).

3.6.2 Electrical preconditioning (burn-in, or debugging). (See 4.6.2). Prior to submission to the specified quality conformance inspections each Regenerative Repeater, MX-9331( )/URC and auxiliary equipments, plus any spare assemblies defined by the contract, shall be electrically preconditioned for a minimum of 48 hours of operating time. The last 24 hours of electrical preconditioning shall be failure free. Failures occurring during this period may be considered non-relevant, but shall be recorded and reported. Burn-in time for all items of equipment submitted for production acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the First Article samples. Test levels and test conditions for burn-in shall be the same as specified in 4.11.2 for the production reliability acceptance test.

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3.7 Service conditions. The equipment shall meet the following service conditions.

3.7.1 Temperature-altitude. (See 4.8.1). The equipment shall be capable of continuous operation over a temperature range of-54°C (-65°F) to +55°C (+131°F) with intermittent operation to +71°C (+160°F), and under combined temperature-altitude conditions in accordance with curves A and B, Figure 3 (sheet 1) of MIL-E-5400 at altitudes from sea level to 30,000 feet. In addition, the equipment shall withstand temperature storage over a range of -62°F (-80°F) to +85°C (+185°F). During testing as specified in 4.8.1, the equipment shall meet the electrical performance requirements of 3.8.2.6 both during and after testing.

3.7.2 Humidity. (See 4.8.2). The equipment shall exhibit no physical damage, such as corrosion, rust, blistering, swelling, or deterioration of parts and materials, and shall meet the requirements of 3.8.2.6 during the operating times specified in 4.8.2.

3.7.3 Vibration, bounce and shock.

3.7.3.1 Vibration. (See 4.8.3). There shall be no physical damage and the equipment shall meet the electrical performance requirements of 3.8.2.6 both during and after testing as specified in 4.8.3.

3.7.3.2 Bounce and shock. The equipment shall be capable of meeting the requirements of Table I.

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TABLE I - BOUNCE AND SHOCK

Paragraph Heading	Paragraph	Performance After Test
Bounce (loose cargo)	4.8.4	Specified performance (NOTE A). No physical damage.
Shock (Ballistic)	4.8.5	Specified performance (Note A). Any physical damage shall be minor only, as classified in MIL-STD-252.
Shock (Bench Handling)	4.8.6	Specified performance (NOTE A). No physical damage.
Shock (Drop)	4.8.7	Specified performance (NOTE A). Any physical damage shall be minor only, as specified in MIL-STD-252.

NOTE A: The equipment shall meet the electrical performance requirements of 3.8.2.6.

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3.7.4 Crash safety. (See 4.8.14). With excursion stops or bumpers in place and with maximum rated load applied in a normal manner, the mounting base, individual isolators, or other attaching devices shall withstand at least 12 impact shocks of 30g, consisting of two shocks in opposite directions along each of three mutually perpendicular axes. Each shock impulse shall have a time duration of 11 ±1 milliseconds. The "g" value shall be within ±10 percent and maximum "g" value shall occur at approximately 5-1/2 milliseconds. Bending and distortion is permitted; however, there shall be no failure of the attaching joints and the equipment or dummy load shall remain in place.

3.7.5 Acceleration. (See 4.8.8). The equipment shall meet the electrical performance requirements of 3.8.2.6 and shall sustain no damage when tested as specified in 4.8.8.

3.7.6 Leakage (immersion). (See 4.8.10). The MX-9331( )/URC and its auxiliary equipments shall show no evidence of leakage after they are tested as specified in 4.8.10.

3.7.7 Dust. (See 4.8.11). The equipment shall meet the electrical performance requirements of 3.8.2.6 after testing as specified in 4.8.11.

3.7.8 Salt fog. (See 4.8.12). After testing as specified in 4.8.12, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

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3.7.9 Fungus. (See 4.8.13). After the fungus test of 4.8.13 the equipment (including accessories) shall be visually examined, using a 10-power magnifier. The equipment (including assessories) shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.5) as evidenced by growth colonization (see 6.5) which includes branching (see 6.5) and sporulation (see 6.5) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.5) shall not be included in this evaluation.

3.7.10 Explosion. (See 4.8.9). The equipment shall not cause ignition of an ambient explosive mixture with air, when operating and tested as specified in 4.8.9.

3.7.11 Operational inspection. (See 4.19). The operational test specified in 4.19 shall be performed to determine that the equipment is operable prior to packaging, and at such other times as required.

3.8 Electrical performance.

3.8.1 Plain mode signal processing. (See 4.12.4).

3.8.1.1 Plain mode switching and monitoring. (See 4.12.4.1 and 4.12.4.2). The MX-9331( )/URC shall pass traffic in the plain mode and automatically switch the radio equipment to the proper mode for retransmission, and shall allow monitoring the traffic using a Handset, H-189/U.

3.8.1.2 Plain mode break-in with or without speech security equipment. (See 4.12.4.3 and 4.12.4.4). The MX-9331( )/URC shall break-in in the direction selected with the function switch using a Handset, H-189/U.

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3.8.2 Cipher mode signal processing. (See 4.12.5).

3.8.2.1 Cipher mode switching and monitoring. (See 4.12.5.1 and 4.12.5.2). The MX-9331( )/URC shall pass traffic in cipher mode at 16.0 or 18.75 kbps, either conditioned baseband or conditioned diphase and automatically switch the radio equipment to the proper mode for retransmission, and shall allow monitoring traffic in the cipher mode with speech security equipment using a Handset, H-189/U.

3.8.2.2 Cipher mode break-in. (See 4.12.5.3 and 4.12.5.4). The MX-9331( )/URC shall break-in in the direction selected with the function switch with speech security equipment in the cipher mode using a Handset, H-189/U, or equal, when operating with radio equipment and/or wireline adapters.

3.8.2.3 Cipher switching, with one wireline adapter. (See 4.12.10). The MX-9331( )/URC with one wireline adapter shall pass traffic in cipher mode conditioned diphase between a 2-wire or 4-wire field wire line and radio equipment and automatically switch the radio equipment and wireline adapter to the proper mode for retransmission.

3.8.2.4 Cipher switching, with two wireline adapters. (See 4.12.11). The MX-9331( )/URC with two wireline adapters shall pass traffic in cipher mode conditioned diphase in both directions between two field wire lines, either 2-wire or 4-wire configuration and automatically switch the wireline adapters to the proper mode for retransmission.

3.8.2.5 Cipher mode translation. (See 4.12.5). The MX-9331( )/URC shall recondition (retime, reshape, reformat) the received cipher mode signals at 16.0 or 18.75 kbps, either conditioned baseband or conditioned diphase and shall be capable of retransmitting the cipher signals in the same mode as received and shall be capable of translating an input conditioned baseband signal to an output conditioned diphase signal and vice-versa, by selecting the proper programming jumpers in the cables interfacing with the radio equipment.

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3.8.2.6 Signal-to-noise ratio. (See 4.12.6). The MX-9331( )/URC shall be capable of regenerating cipher signals with less than I0 percent error for an input signal with a -2 dB signal-to-nosie ratio.

3.8.2.7 Jitter. (See 4.12.9). The cipher mode output signal of the MX-9331( )/URC shall have a jitter of less than 10 microseconds when compared to the input signal.

3.8.2.8 Acquisition time. (See 4.12.7). The time from when a cipher signal is impressed at the input of the MX-9331( )/URC and when a regenerated output is observed shall be less than 35 milliseconds.

3.8.2.9 Release time. (See 4.12.8). The input recovery and retiming circuits of the MX-9331( )/URC shall return to the standby in not less than 100 milliseconds and no more than 300 milliseconds after the removal of the cipher input signal.

3.8.2.10 Wireline adapter input. (See 4.12.11). With the wireline adapter connected to the MX-9331( )/URC and with a 16.0 kbps or 18.75 kbps conditioned diphase cipher signal of 50 millivolts peak-to-peak amplitude entering the wireline adapter at the XMT/2-wire input, when operating in the 2-wire mode or entering at the RCV input when operating in the 4-wire mode, the regenerated cipher output signal of the MX-9331( )/URC shall contain less than one percent errors.

3.8.2.11 Ground loop isolation. (See 4.12.13). A means shall be provided within the radio interface of the clear text retransmission line which will when properly implemented, isolate all ground loops between equipment forming a system.

3.8.2.12 Inversions. (See 4.12.5.1). Digital signals processed by the equipment shall not experience inversions (180° phase change). Accordingly, a digital signal entering the equipment at J1 shall have the same phase as the processed signal leaving it at J2.

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3.8.3 Power requirements. (See 4.12.12).

3.8.3.1 Battery voltage. (See 4.12.12.1). The MX-9331( )/URC shall meet all specified performance requirements when powered by a battery voltage varying from minimum 20 volts dc to maximum 36 volts dc. Battery BA-1590/U or BA-5590/U will be used to power the equipment.

3.8.3.2 Battery power. (See 4.12.12.1 and 4.12.12.2). The power consumption of the MX-9331( )/URC shall not exceed 1.5 watts with the battery supply between 20 and 32 volts dc, and shall not exceed 1.6 watts with the battery supply between 32 and 36 volts dc. The additional power supplied by the battery with one wireline adapter, operating in transmit mode, connected to the MX-9331( )/URC, shall not exceed 1 watt with the battery voltage between 20 and 36 volts dc.

3.8.3.3 Vehicular power. (See 4.12.12.4). The MX-9331 ( )/URC shall meet all specified performance requirements when connected to the Vehicular Power Adapter (VPA) with the VPA powered from a vehicular power source varying from minimum 22 volts dc to maximum 40 volts dc, with transients occurring during normal and abnormal supply characteristics as specified in MIL-STD-1275.

3.8.3.4 Power for speech security device. (See 4.12.12.4.2) The VPA shall be capable of providing power for the TSEC/KY-38 Speech Security Device, connected to the X-mode connector (J3) of the MX-9331( )/URC, under all conditions of vehicular power source specified in 3.8.3.3.

3.8.3.5 Reverse polarity protection. (See 4.12.12.3 and 4.12.12.4.1). The MX-9331( )/URC and the vehicular power adapter shall not sustain any damage when connected to a battery or vehicular power source of reversed polarity and shall meet all specified performance requirements after the polarity is reversed to normal.

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3.8.4 Electrical performance of printed wiring assemblies (modules). (See 4.12.14). Modules shall be tested to determine compliance with the performance requirement specified in paragraph 4.12.14.

3.8.5 Nuclear survivability. (See 4.13). All contractor furnished mechanical configurations, electronic assemblages, electronic equipment, electronic circuits, and electronic components shall withstand the nuclear environment as specified in MIL-R-49107. The equipment shall meet the operational requirements of Section 3 herein immediately after being subjected to the specified nuclear environments. The equipment shall be energized during exposure.

3.9 Electromagnetic interference.

3.9.1 Electromagnetic compatibility. (See 4.10) The MX-9331( )/URC with its Vehicular Power Adapter shall meet the electromagnetic interference requirements specified in MIL-STD-461. The following requirements of that standard are applicable:

CE01	CS01	RE02	RS02
CE03	CS02	RE02.1	RS03
CE04	CS06		RS03.1
CE05			

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3.9.2 Bonding and grounding. (See 4.10.1) Bonding for equipment current path returns, rf potentials, and shock hazard shall be so installed that expansion, contraction, or movement incident to normal service use and maintenance will not break or loosen the connection. Surface preparation for bonds, and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate areas of contact. The equipment finish shall be restored to its original condition in accordance with MIL-F-14072. The dc resistance of bonds and grounds shall not exceed 2.5 milliohms.

3.9.3 Cables and connectors. All shielded cables shall have the shield terminated in an approved military connector with an EMI backshell to provide bonding of the shield. The use of shielded cables shall be kept to a minimum by the use of alternative suppression techniques such as pin filters. All panel mating surfaces shall be free of non-conductive finishes and provide bonding and grounding with mating connectors and equipment ground plane. The DC bonding resistance for all connectors and panel mounted components shall not exceed 2.5 milli-ohms.

3.10 Quantitative reliability requirements. (See 4.11) The MX-9331( )/URC and the MX-9545( )/VR shall provide a specified (θ_0) mean-time-between-failures (MTBF) of 5000 hours when operating under the following conditions:

Ambient temperature: -50°F to +160°F

Duty cycle: 90% retransmission mode, 10% break-in mode

Environment: Airborne

The required reliability shall be demonstrated in accordance with 4.11 of this specification.

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3.11 Interchangeability. (See 4.9) Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.12 Systems safety engineering. (See 4.16) Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.13 Battery case vent. (See 4.4) The battery case shall be provided with a pressure activated vent, set to open at an internal case overpressure of 0.2 psi and in conformance with the requirements of MIL-V-55341.

3.14 Air-seal test, provision for. (See 4.7) The MX-9331( )/URC, MX-9545( )/VR, and CY-7499( )/U enclosure shall provide means for enabling performance of the air-seal test.

3.15 Maintainability. (See 4.15) Quantitative requirement. The following shall be met:

	Mean-Time-To-Repair (MTTR)	Max-Time-To-Repair
Organizational Level Maintenance	10	15
DS/GS Level Maintenance	30	60
Depot Level Maintenance	60	120

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3.16 Weight requirements. Weights of the units shall be as follows:

Unit	Weight not to exceed lbs
MX-9331( )/URC with empty CY-7499( )/U attached	6
MX-9545( )/VR	5

3.17 Workmanship. (See 4.16) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraph:

In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on wire and cable.
Soldering and brazing.
Cleaning.
General requirements for securing of parts.
Cleaning prior to welding.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include preparation for delivery.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging. (See 4.5).

(2) Quality conformance inspection of packaging. (See 4.17).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 First article units. The contractor shall furnish the number of first article units in accordance with the following:

a. Sixteen (16) each MX-9331( )/URC.

b. Sixteen (16) each MX-9545( )/VR.

c. Sixteen (16) each CY-7499( )/U.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table II, and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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Table II - First Article Inspection

Inspections a/	Reqt Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6 Thru 16
1. Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units					
2. Bounce preconditioning	3.6.1	4.6.1	Inspections to be performed on all units					
3. Electrical preconditioning	3.6.2	4.6.2	Inspections to be performed on all units					
4. Group A inspection	(See Table III)		Inspections to be performed on all units					
5. Group B inspection	(See Table IV)		Inspections to be performed on all units					
6. Group C inspection b/, c/								
Weight	3.16	4.14	1				1	
Temperature-altitude	3.7.1	4.8.1	2					
Humidity	3.7.2	4.8.2	3					
Vibration	3.7.3.1	4.8.3		2	3			
Bounce, loose cargo	3.7.3.2	4.8.4		3				
Shock, ballistic	3.7.3.2	4.8.5			4			
Shock, bench handling	3.7.3.2	4.8.6		4				

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Table II - First Article Inspection - Continued

Inspections a/	Reqt Para.	Test Para.	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6 Thru 16
6. Group C inspection b/, c/ (Cont'd)								
Shock, drop	3.7.3.2	4.8.7		5				
Acceleration	3.7.5	4.8.8					3	
Explosive atmosphere	3.7.10	4.8.9				1		
Leakage (immersion)	3.7.6	4.8.10		1&6	2&6			
Dust (fine sand)	3.7.7	4.8.11			1			
Salt Fog	3.7.8	4.8.12					4	
Fungus d/	3.7.9	4.8.13				2		
Shock, crash safety	3.7.4	4.8.14			5			
EMI	3.9	4.10					2	
Nuclear survivability e/	3.8.5	4.13						
7. Group D reliability b/, c/	3.10	4.11						1

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NOTES:

a/ The inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b/ The implementation of inspections covered by 6 and 7 shall be expedited to satisfy contract delivery requirements.

c/ The numbers in the unit columns in the table covered by 6 and 7 specify the order of inspections for the indicated unit.

d/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after the fungus test before proceeding with subsequent inspection.

e/ After completion of Group A and Group B tests, one unit shall be subjected to the nuclear survivability test of paragraph 4.13.13 to demonstrate compliance with the requirements of paragraph 3.8.5.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Parts, materials, and processes, general	3.3
Finish, protective	3.4
Marking	3.5
Battery case vent	3.13
Workmanship	3.17

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4, 4.6, 4.5.1 through 4.5.4 and 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Subassemblies and completed equipments shall be inspected for conformance to the inspections specified in Table III. Each subassembly, after 100% inspection, shall be formed into discrete lots for quality assurance verification. The equipments assembled from lots accepted under subassembly verification shall be inspected 100% and formed into discrete lots for final QA verification. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be submitted to sampling inspection, utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table III (Quality Assurance Verification).

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4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government, except that the operational test in 4.19 shall be next to last and the air-seal in 4.7 shall be last.

Table III - Group A Inspection

Inspection	Reqt. Para.	Insp. Para.	ALQ Major	ALQ Minor
Visual-Mechanical	3.17	4.16		
Printed Wiring Assemblies (Modules)			0.65%	2.5%
Battery Box, CY-7499( )/U			0.65%	2.5%
Case, MX-9331( )/URC			0.65%	2.5%
Case, MX-9545( )/VR			0.65%	2.5%
MX-9331( )/URC, Wired Assembly Less Modules and Case			1.0%	4.0%
Assembled MX-9331( )/URC			1.0%	4.0%
Assembled MX-9545( )/VR			1.0%	4.0%
Electrical				
Printed Wiring Assemblies (Modules)	3.8.4	4.12.14	1.0%	
Air-seal	3.14	4.7	1.0%	
Operational	3.7.11	4.19	1.0%	

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4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and to the special inspection levels of Table 1 of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in Table IV and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

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Table IV - Group B Inspection

Inspection	Reqt. Para.	Test Para.	AQL
Interchangeability			
Dimensional (modules)	3.11	4.9	4.0%
Electrical			
Plain Mode Signal Processing			
Plain mode switching and monitoring	3.8.1.1	4.12.4.1 and 4.12.4.2	
Plain mode break-in	3.8.1.2	4.12.4.3 and 4.12.4.4	
Cipher Mode Signal Processing			
Cipher switching with one wireline adapter	3.8.2.3	4.12.10	
Cipher switching two wireline adapters	3.8.2.4	4.12.11	
Cipher mode translation	3.8.2.5	4.12.5	
Signal-to-noise ratio	3.8.2.6	4.12.6	
Jitter	3.8.2.7	4.12.9	
Acquisition time	3.8.2.8	4.12.7	
Release time	3.8.2.9	4.12.8	
Wireline adapter input	3.8.2.10	4.12.11	
Ground loop isolation	3.8.2.11	4.12.13	
Inversions	3.8.2.12	4.12.5.1	
Power requirements			
Battery voltage	3.8.3.1	4.12.12.1	
Battery power	3.8.3.2	4.12.12.1 and 4.12.12.2	
Vehicular power	3.8.3.3	4.12.12.4	
Power for speech security device	3.8.3.4	4.12.12.4.2	
Reverse polarity protection	3.8.3.5	4.12.12.3 and 4.12.12.4.1	

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4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspection specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

Table V - Group C Inspection

Inspection	Req.Para.	Insp. Para.
Subgroup 1		
Temperature-altitude	3.7.1	4.8.1
Vibration	3.7.3.1	4.8.3
Leakage (immersion)	3.7.6	4.8.10
Subgroup 2		
Humidity	3.7.2	4.8.2
Bounce, loose cargo	3.7.3.2	4.8.4
Subgroup 3		
Shock, ballistic	3.7.3.2	4.8.5
Shock, drop	3.7.3.2	4.8.7
Shock, bench handling	3.7.3.2	4.8.6
EMI	3.9	4.10
Shock, crash safety	3.7.4	4.8.14
Acceleration	3.7.5	4.8.8
Weight	3.16	4.14
Subgroup 4		
Explosive atmosphere	3.7.10	4.8.9
Dust (fine sand)	3.7.7	4.8.11
Salt Fog*	3.7.8	4.8.12
Fungus*	3.7.9	4.8.13

*The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

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4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be selected every month. Both units shall be subjected to leakage (immersion), temperature-altitude, and vibration in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, three units shall be selected at random from the first production lot. For subsequent Group C inspection, three units shall be selected every two months. One unit shall be subjected to humidity testing, one unit shall be subjected to bounce, loose cargo testing, and one unit subjected to acceleration testing.

4.5.3.1.3 Subgroup 3. For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be selected every third month. All tests of this subgroup need not be performed on each unit. The testing may be divided between the two units as long as all subgroup tests are performed.

4.5.3.1.4 Subgroup 4. For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be selected every six months. The tests in this subgroup may be divided between the two units.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

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4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5 Group D inspection. This inspection shall consist of the tests specified in Table VI, and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

4.5.5.1 Sampling for Group D inspection. For reliability purposes a lot shall be defined as one months production.

Table VI - Group D Inspection

Inspection	Req. Para.	Insp. Para.
Reliability	3.10	4.11

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract.

4.5.5.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

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4.6 Preconditioning.

4.6.1 Bounce preconditioning (see 3.6.1). The MX-9331( )/URC with battery box attached, and the Vehicular Power Adapter, shall be placed individually in normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch thick strip of material under one corner or edge of each unit to a distance of two inches as the unit bounces. Each unit shall be subjected to this preconditioning for one minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to the subjection to quality conformance inspection.

4.6.2 Electrical preconditioning (burn-in debugging) (see 3.6.2). Each unit of MX-9331( )/URC and MX-9545( )/VR shall be subjected to the test conditions cited in 4.11.3 and shall meet the requirements of 3.6.2.

4.7 Air-seal test (see 3.14). The MX-9331( )/URC, MX-9545 ( )/VR, and CY-7499( )/U (the battery box shall be checked when attached to the MX-9331( )/URC) shall be opened and closed again in such a manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the equipment enclosure when no transit case is provided. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1 minute period the decrease in vacuum shall not exceed 0.01 pound. The gauge used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily.

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4.8 Service conditions tests. The following tests shall be performed on the equipment (see 3.7).

4.8.1 Temperature-altitude test (see 3.7.1). The equipment shall be tested in accordance with Method 504.1 of MIL-STD-810, for category 4 equipments. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8.2.6. During step 14 of Table 504.1-II, the equipment shall be subjected for two hours to an altitude of 40,000 feet.

4.8.2 Humidity test (moisture-resistance) test (see 3.7.2). The equipment shall be tested in accordance with Method 507.1, Procedure III, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.8.2.6. Measurements shall be made during the last five hours of the last cycle of step 6, and during the last five hours of step 8. Do not move the equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during step 4, it shall be realigned or readjusted and then the test continued. No repair or replacement of parts shall be permitted from step five to the end of the test. If the equipment fails to meet the specified performance during cycling or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

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4.8.3 Vibration test (see 3.7.3.1). The MX-9331( )/URC and vehicular power adapter shall be tested in accordance with the following test methods of MIL-STD-810:

a. Method 514.2, Procedure I for Category b.1 with the equipment mounted on a Government furnished Mounting Base MT-4626( )/URC (with vibration isolators in place, if any). Test level shall be in accordance with Curve D, Figure 514.2-2. In addition, equipment with vibration isolators blocked or removed shall also be tested in accordance with Part 2, Procedure I. The test level shall be according to Curve B, Figure 514.2-2. The time schedules shall be in accordance with Table 514.2-IIA. The equipment shall be operated prior to, during and following vibration testing.

b. Method 514.2, Procedure I for Category C with the equipment mounted on a Government furnished Mounting Base MT-4626( )/URC (with vibration isolators in place, if any). Select test curve from Table 514.2-IIIB, except that Curve M shall be limited to 500 Hz. In addition, equipment with vibration isolators blocked or removed shall also be tested in accordance with Part 2, Procedure I. The test level shall be according to Curve B, Figure 514.2-3. The time schedules shall be in accordance with Tables 514.2-IIIA, except that Curve M shall be limited to 500 Hz and a sweep time of 30 minutes. The equipment shall be operated prior to, during and following vibration testing.

4.8.4 Bounce test (loose cargo) (see 3.7.3.2). The equipment shall be tested in accordance with Method 514.2, Procedure XI, Part 2 of MIL-STD-810.

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4.8.5 Shock test (ballistic) (see 3.7.3.2). The test shall be conducted on the "Shock Testing Machine for Lightweight Equipment" shown in Figure 1 of MIL-S-901. The MX-9331( )/URC with its Vehicular Power Adapter to be tested shall be mounted on a Government-furnished Mounting Base MT-4626( )/URC. The equipment with its Mounting Base shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of nine blows; one each 1-foot blow, 3-foot blow, and a 5-foot blow on the back, side, and top of the test plate. As an alternate to re-orienting the plate for the blows on the side of the plate, an equivalent rotation of the equipment under test is permissible.

4.8.6 Shock test (bench handling) (see 3.7.3.2). The equipment shall be tested in accordance with Method 516.2, Procedure V, of MIL-STD-810.

4.8.7 Drop test (see 3.7.3.2). The equipment shall be tested in accordance with Method 516.2, Procedure II, of MIL-STD-810.

4.8.8 Acceleration (see 3.7.5). The MX-9331( )/URC as normally mounted shall be tested according to MIL-STD-810, Method 513.2, Procedure II. The test level shall be 6g with 1 minute duration in each direction of each axis for a total test time of 6 minutes.

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4.8.9 Explosion (see 3.7.10). The MX-9331( )/URC shall be tested according to Method 511.1, Procedure I of MIL-STD-810. Test altitude shall be 30,000 feet maximum.

4.8.10 Leakage test (immersion) (see 3.7.6). The equipment shall be tested in accordance with Method 512.1, Procedure I, of MIL-STD-810. This test shall be performed prior to and following vibration and shock testing of the same test item.

4.8.11 Dust test (see 3.7.7). The equipment shall be tested in accordance with Method 510.1, Procedure I, of MIL-STD-810, except that the equipment shall be operating for one hour of each 6-hour period of steps 1 and 3.

4.8.12 Salt fog test (see 3.7.8). The equipment shall be subjected to the test of Method 509, Procedure I, of MIL-STD-810. Electronic operation shall not be required and failure criteria shall be limited to the following:

a. Corrosion of base metal or metallic surface finishes; corrosion being defined as any visible degradation (as seen with the unaided eye except for eyeglasses) of equipment internal or external surfaces such as flaky, pitted, blistered, delaminated or otherwise loosened finish or base metal.

b. Mechanical operation of moving parts such as exposed bearing surfaces, switches, gearing, etc., shall be conducted through the full range of movement or adjustment without clogging or binding of mating parts or surfaces.

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4.8.13 Fungus test (see 3.7.9). The equipment shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810. For gasket sealed or other non-hermetic sealed equipments, the parts and materials of the chassis shall be sprayed with spore suspension. The interior and exterior of the case shall also be sprayed. The equipment shall remain in an open condition during the test. Operation is not required. Inability of the equipment to meet the requirements of 3.7.9 shall constitute failure of the test.

4.8.14 Crash safety (see 3.7.4). The equipment mounted on a Government-furnished Mounting Base MT-4626( )/URC shall be subjected to the test of Method 516.2, Proceudre III of MIL-STD-810. The shock pulses shall be in accordance with Figure 516.2-2 for flight vehicle equipment. Test shall be performed on MX-9331( )/URC with MX-9545( )/VR and CY-7499/U (with Battery BA-1590/U in battery case) attached.

4.9 Inspection for dimensional interchangeability (see 3.11). The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. MX-9331( )/URC

(1) The maximum dimensions of chassis, interface with case, and location of four (4) mounting holes.

(2) The minimum inside dimensions of case and location of four (4) mounting holes.

(3) The each of four (4) pluck-out boards for maximum dimensions, maximum component heights, guide pins, keying, and height and location of connector.

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4. The each of four (4) cavities for above boards for minimum dimensions, spacing, location of slides, guide bushings, keying, and location and height of connector.

5. The interface dimensions with the battery box at back of case including connector height and location.

b. CY-7499/U

1. The battery box for interface dimensions with the MX-9331 repeater including connector height and location, and the minimum inside box dimensions.

4.10 Electromagnetic compatibility (see 3.9.1). The MX-9331( )/URC shall be tested for compliance with the requirements of 3.9.1. Test setups and procedures shall be in accordance with MIL-STD-462, Notice 3, as implemented by a contractor prepared, Government approved, EMI test plan. Signal and data input and output terminals shall be terminated with simulated inputs and loads. Simulated loads shall have characteristic impedance similar to actual loads. All cables used during test shall be supplied by the contractor and of the same type and length used in the actual installation. All electromagnetic compatibility tests shall be performed with the vehicular power supply installed.

4.10.1 Bonding impedance (see 3.9.2 & 3.9.3). Prior to the performance of EMI/EMC testing of 4.10, bonding impedances shall be measured for compliance to the requirements of 3.9.2 and 3.9.3. All measurement readings shall be recorded and included in the EMI/EMC test report. Measurements shall be performed from the equipment case to the shell of case mounted connectors and all other externally mounted connectors. Additionally, measurements shall be performed between all mating surfaces bonded by EMI gasketing.

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4.11 Reliability (see 3.10). The following reliability tests shall be performed on the equipment.

4.11.1 Reliability qualification test. The contractor shall perform this test as part of first article testing. The reliability of the equipment shall be demonstrated by selecting ten First Article samples of MX-9331( )/URC and MX-9545( )/VR that have passed Group A and Group B inspection and subjecting them to the requirements of Test Plan XXII of MIL-STD-781. They shall demonstrate a specified MTBF of 5000 hours. Test levels and test conditions shall be as specified in paragraph 4.11.3. No single equipment "on" time shall be less than one-half the average operating time of all equipments on test.

4.11.2 Production reliability sampling test. The reliability of production equipment shall be demonstrated in accordance with Test Plan V of MIL-STD-781. A maximum of ten samples that have passed Group A and Group B inspection shall be used. The specified MTBF is 5000 hours. Test levels and test conditions shall be as specified in paragraph 4.11.3. No single equipment "on" time shall be less than one-half the average operating time of all equipment on test.

4.11.2.1 Sampling. Ten samples shall be selected at random once each fourth lot or 140 units, whichever is sooner.

4.11.3 Test conditions.

4.11.3.1 Cycle. The testing shall be conducted on a 24-hour, 7-day-per-week basis.

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4.11.3.2 Temperature cycling. Temperature cycling shall be in accordance with the standard method of temperature cycling of MIL-STD-781.

4.11.3.3 Environmental and operation stresses. Vibration and temperature cycling requirements shall be as specified in Test Level F of MIL-STD-781 except the low temperature test level shall be -45.5°C. Equipment ON-OFF cycling and input voltage cycling are not applicable.

4.11.3.4 Performance parameters to be measured. Each unit shall meet all the electrical requirements specified in paragraph 3.8.2.6 at low temperature, high temperature, and room temperature. Testing shall be performed at each of these temperatures on a once a day basis as follows: Test 4.12.6 shall be performed at temperature extremes after stabilization, measure errors one time in each signal direction; tests 4.12.4, 4.12.5.1 or 2, 4.12.5.3 or 4, shall be performed at room ambient temperature. Each unit shall operate for half its test time at 16.0 kbps and for the other half of test time at 18.75 kbps during the Reliability Qualification Test. The production reliability sampling test shall test the units at the bit rate frequency set as required for shipment (see 6.2.h).

4.11.4 Reliability failures. The following requirements apply to both the Reliability Qualification Test and the Production Reliability Sampling Test.

4.11.4.1 Failure actions, categories and confirmation. Failure actions, categories and confirmation shall be as specified in MIL-STD-781.

4.11.4.2 Verifying repair. Post repair verification shall be as specified in MIL-STD-781.

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4.11.4.3 Preventive maintenance. No preventive maintenance shall be performed on the units during the Realiability Test. However, operator controls shall be adjusted as required.

4.11.4.4 Test reports and failure reporting and analysis. Test reports, failure reporting and failure analysis shall be as specified in the contract.

4.12 Electrical tests.

4.12.1 Standard conditions. Unless otherwise specified, all tests shall be performed with a nominal 24 Vdc power supply or Battery BA-1590 or BA-5590, and shall be conducted at room ambient temperature, pressure and humidity.

4.12.2 Special test equipment. Unless otherwise specified, special test equipment and interconnecting cables shall be provided by the contractor subject to approval of the Government.

a. Test Set A, Operational Test Set, see Figure 1.

b. Test Set B, Signal/Noise Test Set, see Figure 2.

c. Test Set C, Error Counter Test Set, see Figure 3.

4.12.3 Performance tests. All tests are conducted with Test Sets A, B, and C. Interconnection of cables and test equipment shall be as shown in Figures 1 and 4.

4.12.4 Plain mode signal processing.

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4.12.4.1 Plain mode retransmission and monitoring (see 3.8.1.1). With the MX-9331( )/URC function switch in RXMT position, connect Test Set A (see Figure 1) to MX-9331( )/URC as shown in Figure 4. Apply a 1000 Hz, 2.7 ±0.1 V rms audio signal to RT1 input (J1-B) and connect RT1 retrans (J1-K) to ground. Verify presence of audio signal of 13.5 ±2 mV rms at RT2 output (J2-D). Verify proper indication of RT2 indicator lamps.

Tone disable	OFF
PTT	ON

Verify presence of audio signal at X-mode (J3-B) and handset (J4-B) outputs, 2.7 ±0.25V rms with volume control full clockwise and 0.6 ±0.1V rms with volume control at 50 percent of rotation.

4.12.4.2 Repeat 4.12.4.1 with exchange of RT1 and RT2 (see 3.8.1.1).

4.12.4.3 Plain mode break-in (see 3.8.1.2).

a. With the MX-9331( )/URC function switch in RT1 position, apply signal and retrans to RT1 as in 4.12.4.1. Verify presence of signal at X-mode and handset outputs as in 4.12.4.1.

b. Apply audio signal of 1000 Hz, 0.4 ±0.01V rms, to microphone inputs of X-mode (J3-D) and handset (J4-D) connector. Verify presence of audio signal at RT1 output (J1-D), 0.4 ±0.04V rms. Verify proper indication of RT1 indicator lamps

Tone disable	ON
PTT	OFF

Switch X-mode PTT (J3-C) and handset PTT (J4-C) to ground, one at a time, and verify that RT1 indicator lamp show

Tone disable	OFF
PTT	ON

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4.12.4.4 Repeat 4.12.4.3 with exchange of RT1 and RT2 (see 3.8.1.2).

4.12.5 Cipher mode signal processing (see 3.8.2). Operate MX-9331( )/URC at bit rate of 18.75 kbps (jumper E1-E3 on A2).

NOTE: The cipher mode output signals of the MX-9331( )/URC appear inverted or non-inverted as compared to the input signal, depending on setting of baseband/diphase programming switch of Test A, both signals are equally valid.

4.12.5.1 Cipher mode retransmission and monitoring at 18.75 kbps bit rate (see 3.8.2.1 and 3.8.2.12).

a. With the MX-9331( )/URC function switch in the RXMT position, apply the 8-bit conditioned baseband signal of Test Set A (see Figure 1) at bit rate of 18.75 kbps, 9 ±1V peak to peak, to RT1 input (J1-P). Verify presence of the same conditioned baseband cipher signal (see Figure 5) at RT2 output (J2-R), at 4 ±1V peak to peak into 1.8-kohm load. Verify proper operation of indicator lamp. Verify that the phase of the signal emanating from (J1-R) is the same as that applied at (J2-R).

RT1 Tone disable	ON
RT1 PTT	OFF
RT2 Tone disable	ON
RT2 PTT	ON

b. Verify presence of conditioned baseband cipher signal at X-mode (J3-P) output, 10 ±2V peak to peak (Figure 7) and at handset (J4-B) monitoring output, 6 ±1V peak to peak in 2.0-kohm load (Figure 8) with volume control full clockwise.

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c. Set baseband/diphase programming switch on Test Set A to diphase. Verify presence of conditioned diphase cipher signal (see Figure 6) at RT2 output (J2-R) at 7 ±1.5V peak to peak, into 1.8-kilohm load.

d. Verify that signals of 4.12.5.1b remained in conditioned baseband mode at same peak-to-peak levels.

e. Set baseband/diphase selection switch on Test Set A to diphase. Set Test Set A output level to 1 ±0.5V peak to peak. Verify presence of conditioned diphase cipher signal (see Figure 6) at RT2 output (J2-R), at 7 ±1.5V peak to peak, into 1.8-kilohm load.

f. Repeat 4.12.5.1d.

g. Set baseband/diphase programming switch of Test Set A to baseband. Verify presence of conditioned baseband cipher signal (see Figure 5) at RT2 output (J2-R), at 4 ±1V peak to peak into 1.8-kilohm load.

h. Repeat 4.12.5.1d.

i. Repeat 4.12.5.1a through 4.12.5.1h with exchange of RT1 and RT2.

4.12.5.2 Cipher mode retransmission and monitoring at 16.0 kbps bit rate (see 3.8.2.1). Operate MX-9331( )/URC at bit rate of 16.0 kbps (jumper E2-E3 on A2).

a. With Test Set A operating at 16.0 kbps bit rate, repeat 4.12.5.1a through 4.12.5.1h.

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b. With Test Set A operating at 16.0 kbps bit rate, repeat 4.12.5.1j.

4.12.5.3 Cipher mode break-in at 16.0 kbps bit rate (see 3.8.2.2). Continue to operate MX-9331( )/URC at bit rate of 16.0 kbps.

a. With the MX-9331( )/URC function switch in RT1 position, apply the 8-bit conditioned baseband signal of Test Set A at 16.0 kbps, 9 ±1V peak to peak, to RT1 input (J1-P). Set baseband/diphase programming switch on Test Set A to baseband. Repeat 4.12.5.1b.

b. Set baseband/diphase selection switch of Test Set A to diphase. Repeat 4.12.5.1d.

c. Set baseband/diphase programming switch of Test Set A to diphase. Repeat 4.12.5.1d.

d. Set baseband/diphase selection switch of Test Set A to baseband. Repeat 4.12.5.1d.

e. With the MX-9331( )/URC function switch in RT1 position, apply the 8-bit conditioned baseband signal of Test Set A at 16.0 kbps, 9 ±1V peak to peak, to X-mode input (J3-R). Set baseband/diphase programming switch of Test Set A to baseband. Verify presence of conditioned baseband cipher signal (see Figure 5) at RT1 output (J1-R), at 4 ±1V peak to peak into a 1.8-kilohm load. Verify proper operation of indicator lamps

RT1 Tone disable	ON
RT1 PTT	ON
RT2 Tone disable	ON
RT2 PTT	OFF

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f. Set baseband/diphase programming switch of Test Set A to diphase. Verify presence of conditioned diphase cipher signal (see Figure 6) at RT1 output (J1-R) at 7 ±1.5V peak to peak into a 1.8-kilohm load.

g. With the MX-9331( )/URC function switch in RT2 position, repeat 4.12.5.3a through 4.12.5.3f with exchange of RT1 and RT2.

4.12.5.4 Cipher mode break-in at 18.75 kbps bit rate (see 3.8.2.2). Operate MX-9331( )/URC at bit rate of 18.75 kbps (jumper E1-E3 on A2).

a. With Test Set A operating at 18.75 kbps bit rate, repeat 4.12.5.3a through 4.12.5.3g.

4.12.6 Signal-to-Noise ratio (see 3.8.2.6).

a. With Test Set A connected to the MX-9331( )/URC, apply the output of Test Set B, consisting of the 127-bit pseudo-random sequence with added noise, to the X-mode input of Test Set A. With the MX-9331( )/URC function switch in RXMT, apply this signal to RT1 input (J1-P). With the clean signal set to 1V peak to peak, add noise to obtain -2dB signal-to-noise ratio. Verify the presence of the reconstructed signal at RT2 (J2-R) output of the MX-9331( )/URC and verify that indicator lamp RT2-PTT is ON. Connect Test Set C to the output from MX-9331( )/URC at Test Set A and synchronize (injection lock) Test Set C to sequence. Count cipher signal errors for 100 seconds. Repeat four times.

b. Repeat with exchange of RT1 and RT2.

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e proper programming jumpers in the vice-versa, by selecting in

4.12.7 Signal acquisition time (see 3.8.2.8).

a. With the MX-9331( )/URC function switch in RXMT, apply the 8-bit sequence from Test Set A to RT1 (J1-P) input and verify the presence of the MX-9331( )/URC output signal at RT2 output. Operate both equipments at a bit rate of 16.0 kbps in conditioned diphase mode. Trigger oscilloscope from channel 1 (input signal) in signal trace triggering mode at 5 millisec/cm sweep speed. Switch Test Set A signal off and on with baseband/off/diphase selection switch. Measure on oscilloscope delay between start of trace and start of signal. Repeat four times.

b. Repeat with exchange of RT1 and RT2.

4.12.8 Signal release time (see 3.8.2.9).

a. Connect equipments as above in 4.12.7a. Connect input A of oscilloscope to the RT2-PTT indicator light test point (TP2) in Test Set A. Trigger the oscilloscope externally from test point TP3 of baseband/off/diphase selection switch in single trace triggering mode at 50 millisec/cm sweep speed. Switch baseband/off/diphase selection switch from diphase to off. Measure on oscilloscope the delay between start of trace and the step on trace 1, which indicates the end of RT2-PTT function.

b. Repeat with exchange of RT1 and RT2.

4.12.9 Signal jitter (see 3.8.2.7).

a. Connect equipments as above in 4.12.7a. Trigger oscilloscope from signal trigger on Test Set A in alternate trigger mode. Observe input and output signal on oscilloscope, increase sweep speed of oscilloscope until horizontal jitter of the output signal can be observed. Determine the amount of horizontal peak-to-peak jitter.

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b. Repeat with exchange of RT1 and RT2.

4.12.10 Wireline adapter output (see 3.8.2.3).

a. With the MX-9331( )/URC connected to Test Set A apply the 8-bit sequence at 16.0 kbps bit rate conditioned diphase signal to RT1 (J1-P) input. Observe input and output signals on oscilloscope. Disconnect cable from the RT2 (J2) connector of the MX-9331( )/URC and connect the Wireline Adapter to RT2. Connect a load resistor of 170 ±5 ohms to terminals XMT/2-WIRE of Wireline Adapter. Connect oscilloscope channel 1 to one terminal and connect channel 2 to the other terminal; set oscilloscope mode switch to ADD and set channel 2 polarity to INVERT. Measure peak-to-peak signal across 170-ohm resistor.

b. Repeat with exchange of RT1 and RT2.

4.12.11 Wireline adapter input (see 3.8.2.4 and 3.8.2.10).

a. Connect Wireline Adapter to RT2 (J2) connector of MX-9331( )/URC. Connect Test Sets A, B, and C to Wireline Adapter and MX-9331( )/URC as shown in Figure 9. Apply signal from Test Set B through transformer T to XMT/2-WIRE input of Wireline Adapter. Set Wireline Adapter switch to 2-WIRE.

b. Observe the reconstructed signal at RT1 (J1) output of the MX9331( )/URC and verify that indicator lamp RT1-PTT is on. Synchronize (injection lock) Test Set C to signal sequence. Count cipher signal errors for 100 seconds. Repeat two times.

c. Apply signal to RCV input of Wireline Adapter and set Wireline Adapter switch to 4-WIRE. Repeat 4.12.11.b.

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d. Repeat 4.12.11a, 4.12.11b, and 4.12.11c with exchange of RT1 and RT2. .

e. With equipments operating as in 4.12.11a, connect a second Wireline Adapter to RT1 (J1) and connect the 170 ±5-ohm load resistor to terminals XMT/2 WIRE of the second Wireline Adapter. Connect oscilloscope and Test Set C as shown in Figure 10. Observe the reconstructed signal at the Wireline Adapter output. Synchronize (injection lock) Test Set C to signal sequence. Count cipher signal errors for 100 seconds.

f. Apply signal from Test Set B to second Wireline Adapter and perform test 4.12.11e on the first Wireline Adapter.

4.12.12 Power supply.

4.12.12.1 Power to MX-9331( )/URC. (See 3.8.3.1 and 3.8.3.2). Connect MX-9331( )/URC to Test Set A and operate in cipher mode retransmission. Observe input and output signals on oscilloscope. Step the dc supply connected to the battery connector of the MX-9331( )/URC through the following voltages: 20, 24, 28, 32 and 36V dc, all ±0.1V. Measure supply current of each step and calculate power consumption in watts. Observe that at each step the input/output signals on oscilloscope remain unchanged.

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4.12.12.2 Power to MX-9331( )/URC with Wireline Adapter. (See 3.8.3.2). Disconnect cable from the RT2 (J2) connector of MX-9331( )/URC and connect Wireline Adapter to RT2. Connect 170 ±5-ohm resistor across Wireline Adapter terminal XMT/2-WIRE. Operate in cipher mode retransmission with Test Set signal applied to RT1. Connect oscilloscope channel 2 to one side of 170-ohm resistor, and observe MX-9331( )/URC input signal and Wireline Adapter output signal on oscilloscope. Step supply voltage as in 4.12.12.1; measure currents and calculate power consumption. Observe that input/output signals remain unchanged at each step.

4.12.12.3 Reversed polarity battery. (See 3.8.3.5) With equipments operating as in 4.12.12.2, set supply voltage to 24 Vdc. Reverse polarity of supply and keep connected to equipment for five minutes. Return polarity to normal and operate equipment. Verify that input/output performances as checked in 4.12.12.1 and 4.12.12.2 remained unchanged.

4.12.12.4 Vehicular power supply. (See 3.8.3.3)

a. Connect Vehicular Power Adapter to MX-9331( )/URC. Apply 27.5 ±0.1 Vdc to Vehicular Power Adapter and operated equipments as in 4.12.12.1. Measure supply current into Vehicular Power Adapter at 22, 27.5 and 32 Vdc input.

b. Connect Wireline Adapter to MX-9331( )/URC and operate equipments as in 4.12.12.2. Measure supply current into Vehicular Power Adapter at 22, 27.5, and 32 Vdc input.

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4.12.12.4.1 Reversed polarity vehicular power. (See 3.8.3.5) With equipments operating as in 4.12.12.4b, set supply voltage to 27.5 VDC. Reverse polarity of supply and keep connected for five minutes. Return polarity to normal and operate equipment. Verify that input/output performance as checked in 4.12.12.1 and 4.12.12.2 remained unchanged.

4.12.12.4.2 Power for speech security device. (See 3.8.3.4) With equipments operating as in 4.12.12.4b, connect a 150-ohm ±5%, 5-watt resistor between pin E of the X-MODE (J3) connector of the MX-9331( )/URC and ground, to simulate the power supplied to the TSEC/KY-38. Operate for 15 minutes. Verify that input/output performance as checked in 4.12.12.1 and 4.12.12.2 remained unchanged.

4.12.12.4.3 Vehicular power transients. (See 3.8.3.3) With the equipments operating as in 4.12.12.4b, apply the abnormal power conditions of MIL-STD-1275 to the input of the Vehicular Power Adapter, consisting of a steady state voltage of 40 Vdc with a transient of 100 Vdc lasting 50 milliseconds and decaying in 200 milliseconds, having a repetition rate of one second, Operate for 15 minutes and observe input/output performance for adverse effects. After completion of test, verify that input/output performance as checked in 4.12.12.1 and 4.12.12.2 remained unchanged.

4.12.13 Ground loop tests. (See 3.8.2.11) With the equipment interconnected to an AN/VRC-49 using CX-12126/U cables, clear text retransmissions shall be performed to determine the quality, SNR, of the retransmitted signal. The signal to noise ratio shall be in excess of 10db under nominal conditions.

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4.12.14 Electrical performance of printed wiring assemblies (modules). (See 3.8.4) Modules shall be checked to determine compliance with the applicable test procedure drawing referenced in drawings covering construction of the modules.

4.13 Nuclear survivability tests. (See 3.8.5) Nuclear tests (blast, thermal, EMP, Initial Nuclear Radiation) shall be performed on the equipment in accordance with MIL-R-49107. The equipment shall meet the operational requirements of section 3 herein following each test.

4.14 Weight. (See 3.16) The equipment shall be weighed to show compliance with 3.16.

4.15 Maintainability. (See 3.15) The maintainability requirements (Mc and M_{MAXCT}) shall be demonstrated in accordance with Test Method 9 of MIL-STD-471, Notice 1, dated 10 Jan 1975 using a minimum sample size of 30 corrective maintenance tasks at each specified level of maintenance. The consumers risk (β) for demonstration of Mc and M_{MAXCT} will be 10%. The specified maximum corrective maintenance downtime (M_{MAXCT}) will be taken to represent to 90 (γ) percentile point of the downtime distribution. The equipment will be considered to have met the requirements when all the specified parameters have received accept decisions. If one or more specified parameters fail to meet the accept decision, then design corrective action must be initiated at the contractor's expense. When proper design corrective action has been completed, the equipment will be retested at those levels of maintenance that previously failed. The same accept/reject criteria will apply in the retest phase.

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4.15.1 Maintainability test conditions. The mean-time-to-repair for organizational, DS/GS and depot level shall meet the requirements of 3.15.

4.16 Visual and mechanical inspection. (See 3.12 and 3.17). The equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those portions of 3.12 (safety engineering) which can be determined visually.

4.17 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

4.18 Rough handling test packaging. When rough handling test in accordance with MIL-P-116 is required by the contract (see 6.2), the electrical operation performance tests (4.19) shall be conducted to determine freedom from operational malfunction caused by rough handling.

4.19 Operational inspection. All equipments to be operationally inspected shall be tested in accordance with 4.12.5.1, 4.12.5.2, 4.12.5.3 and 4.12.5.4.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Regenerative Repeater MX-9331( )/URC and auxiliary equipments, Vehicular Power Adapter MX-9545( )/VR, Battery Box CY-7499( )/U and Attaching Cables CX-12066( )/U shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Regenerative Repeater MX-9331( )/URC and auxiliary equipments, Vehicular Power Adapter MX-9545( )/VR, Battery Box CY-7499( )/U and Attaching Cables CX-12066( )/U shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. Not required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be individually unit packed method IC-1.

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5.1.1.4.2 Regenerative Repeater MX-9331( )/URC, Battery Box CY-7499( )/U and Attaching Cables CX-12066( )/U. Each Regenerative Repeater MX-9331( )/URC with Battery Box CY-7499 ( )/U and Attaching Cables CX-12066( )/U shall be individually unit packed method III as follows: the Battery Box CY-7499 ( )/U shall be mounted to the rear of the repeater and the two units secured together by the two side latches. Two each Attaching Cables CX-12066( )/U shall be individually coiled and secured with twine conforming to T-T-871 at three equidistant places. Individually cushion the repeater-battery box assembly and the attaching cables on all surfaces with cells or pads (or both) fabricated from fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned items within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, variety SW, grade W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1 on top of the contents directly under the box flaps. Close the box in accordance with the appendix of the box specification.

5.1.1.4.3 Vehicular Power Adapter MX-9545( )/VR. Each Vehicular Power Adapter MX-9545( )/VR, shall be individually unit packed method III as follows: Individually cushion each vehicular power adapter on all surfaces with cells or pads (or both) fabricated from fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. The cushioned vehicular or power adapter shall be placed into a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant, variety SW and grade W5c. Close the box in accordance with the appendix of the box specification.

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5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservative application and unit packing. Each Regenerative Repeater MX-9331( )/URC and auxiliary equipments, Vehicular Power Adapter MX-9545( )/VR, Battery Box CY-7499( )/U and Attaching Cables CX-12066( )/U shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.

5.1.3 Commercial. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Regenerative Repeater MX-9331( )/URC with Battery Box CY-7499( )/U and Attaching Cables CX-12066( )/U, or a quantity of Vehicular Power Adapter MX-9545( )/VR, bearing the same stock number and unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers with items bearing the same stock number shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-601, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

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5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.2 Level B. A quantity of Regenerative Repeater MX-9331( )/URC with Battery Box CY-7499( )/U and Attaching Cables CX-12066( )/U, or a quantity of Vehicular Power Adapter MX-9545( )/VR, bearing the same stock number and unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. A quantity of the containers with items bearing the same stock number shall be packed within a close fitting fiberboard box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

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5.2.2.1 Unitization. Palletization shall be required when: containers specified in 5.2.2 do not require pallet mounting; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.2, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be 'bond-d' to the pallet by strapping conformin to QQ-S-781, type 1, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

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6. NOTES

6.1 Intended use. The Regenerative Repeater, MX-9331( )/URC and its auxiliary equipments, is a rugged, lightweight, compact equipment which shall extend the range of Radio Sets when operating in plain voice and with cipher mode of either conditioned baseband or diphase with speech security equipments of the Nestor, and Saville families.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Auxiliary equipments required (see 1.1).

c. Levels of preservation. packaging and packing required for shipment. (see 5.1).

d. First Article required (see 3.2).

e. Marking and shipping of samples.

f. If the rough handling and functional tests are required (see 4.18).

g. Specify quantities of units to be shipped for operation at 16.0 kbps and for operation at 18.75 kbps.

6.3 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design. As soon as possible after the award of contract, the contractor should apply to the Contracting Officer for nomenclature in accordance with the applicable clause of the contract.

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6.4 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provision established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Definitions.

6.5.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.5.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.5.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.5.4 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

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6.5.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Location of operational inspection and air-seal test. It is desirable that the Operational Test (paragraph 4.19) and the Air Seal Test (paragraph 4.7) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the Air-Seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

Custodian:
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Preparing Activity:
Army - EL
Project Number 5810-A033-1

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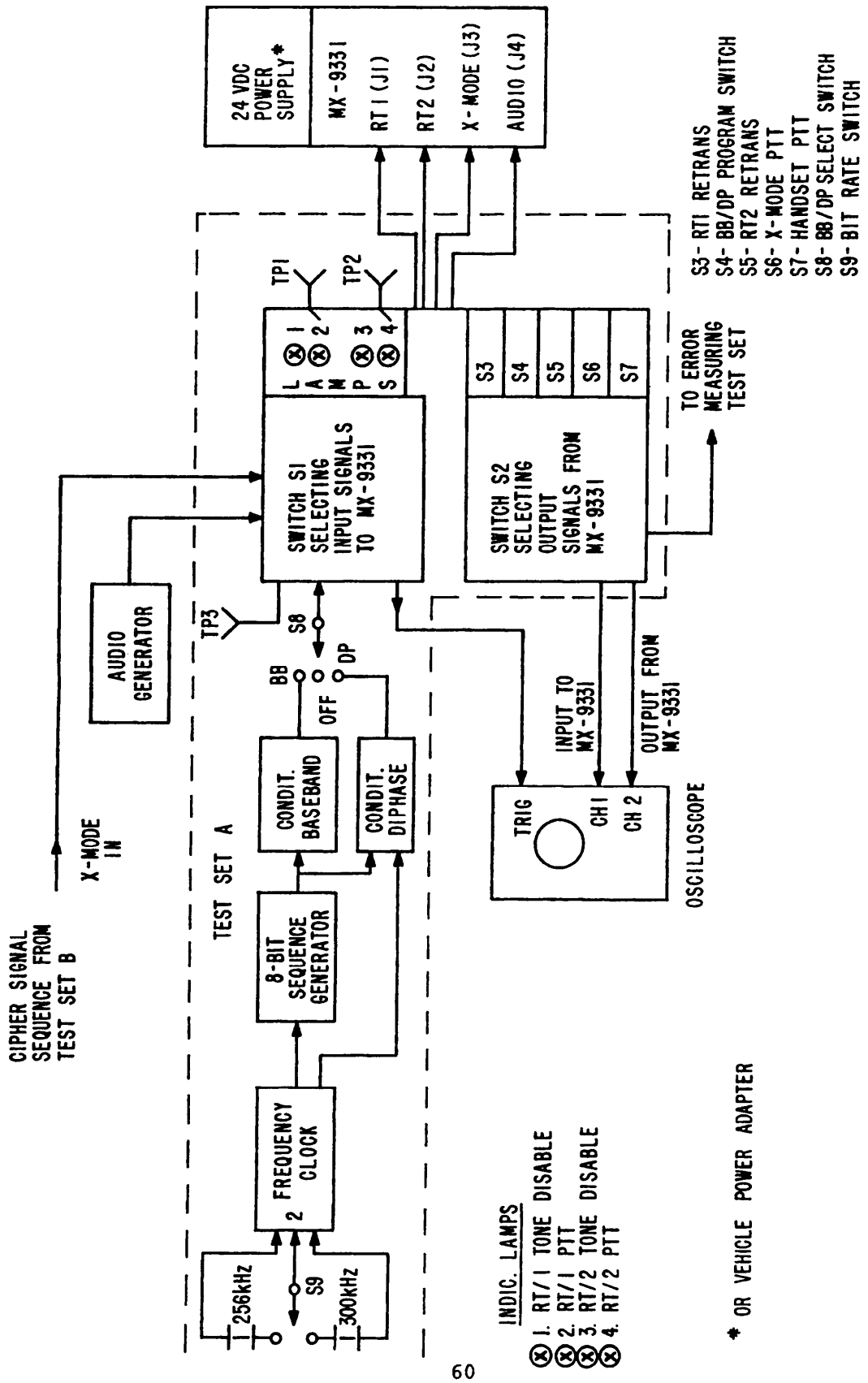


FIGURE 1. BLOCK DIAGRAM OF OPERATIONAL TEST SET A

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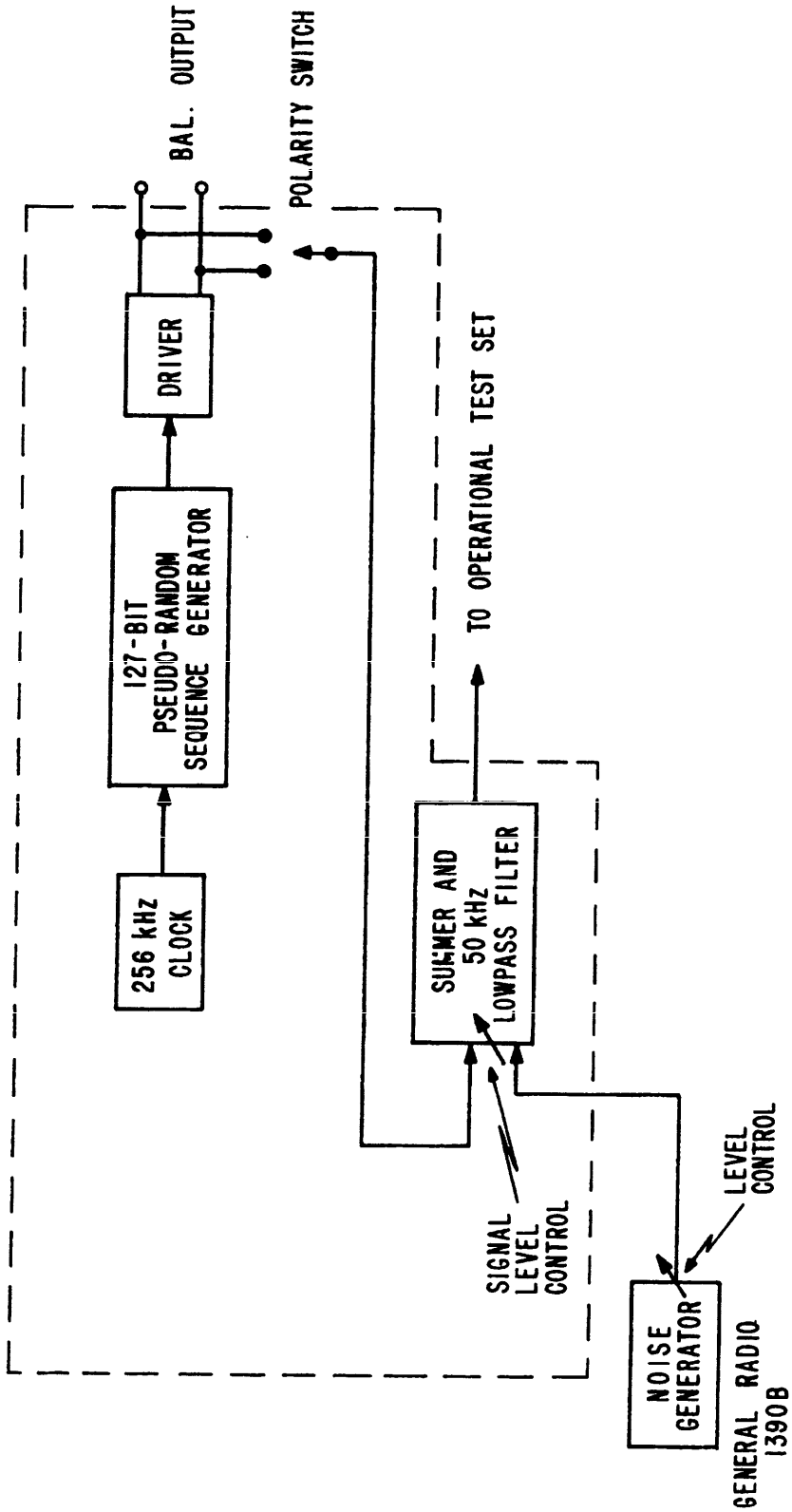


FIGURE 2. SIGNAL/NOISE PERFORMANCE TEST SET B

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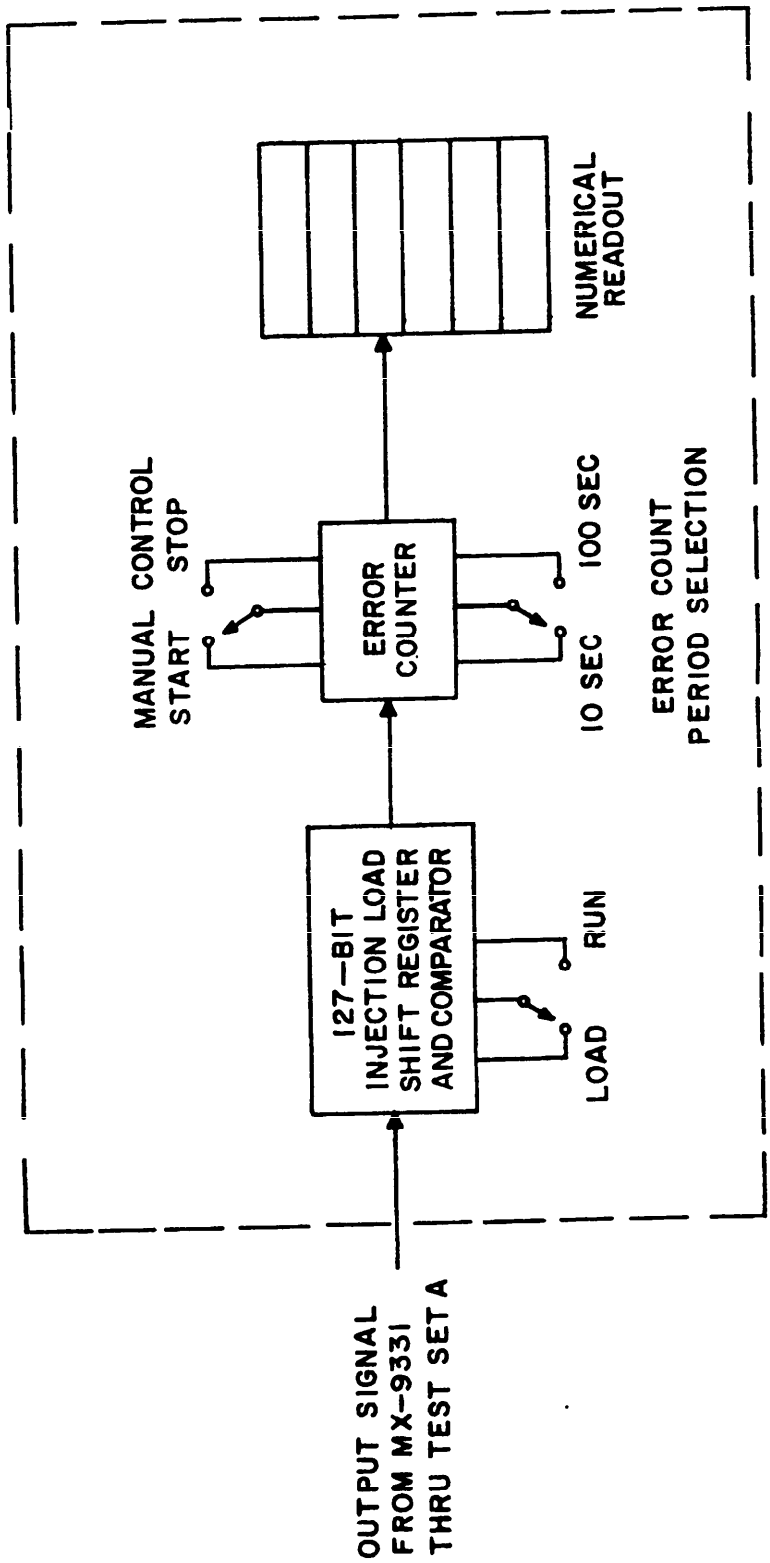


FIGURE 3. COMPARATOR/ERROR COUNTER TEST SET C

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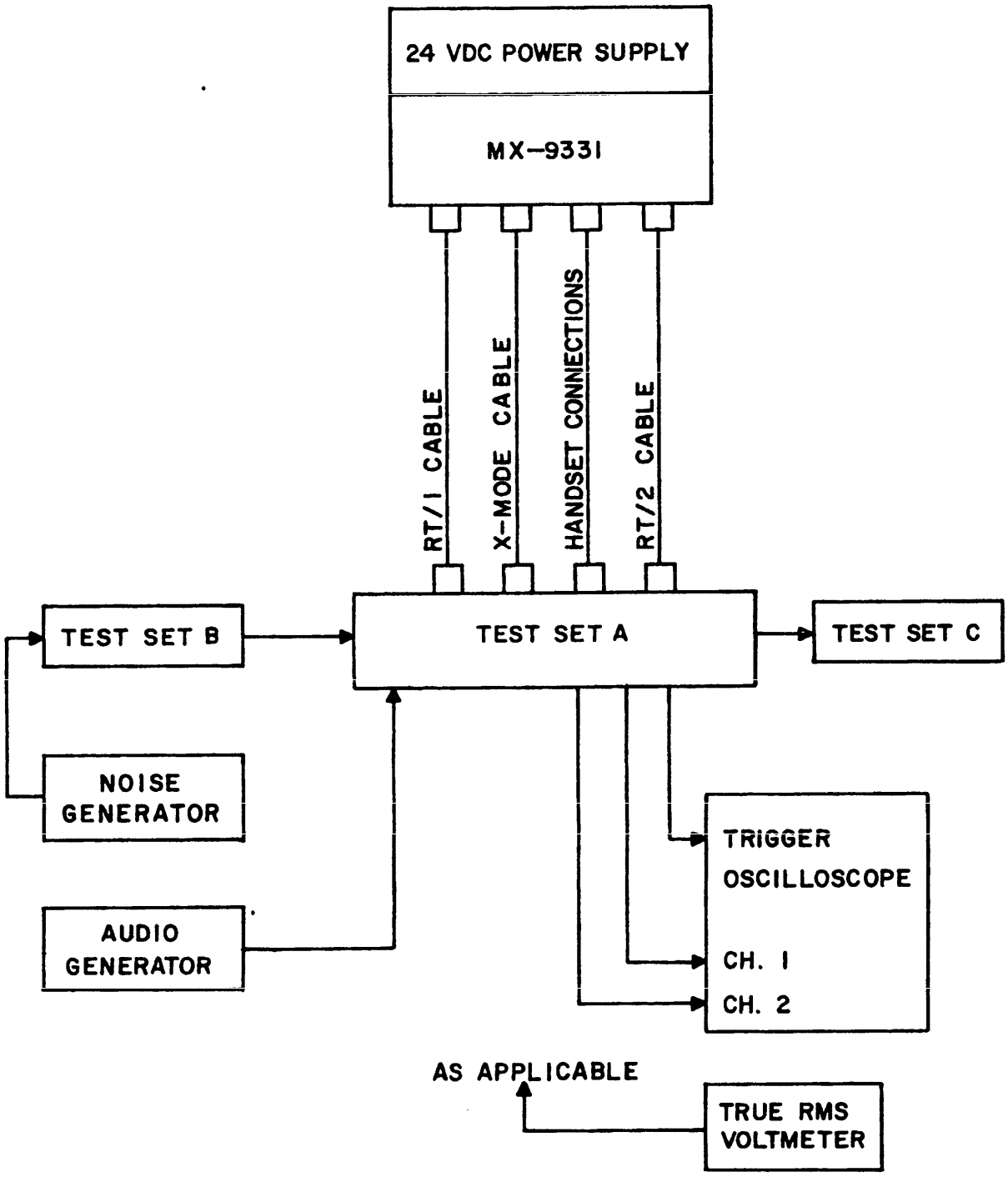
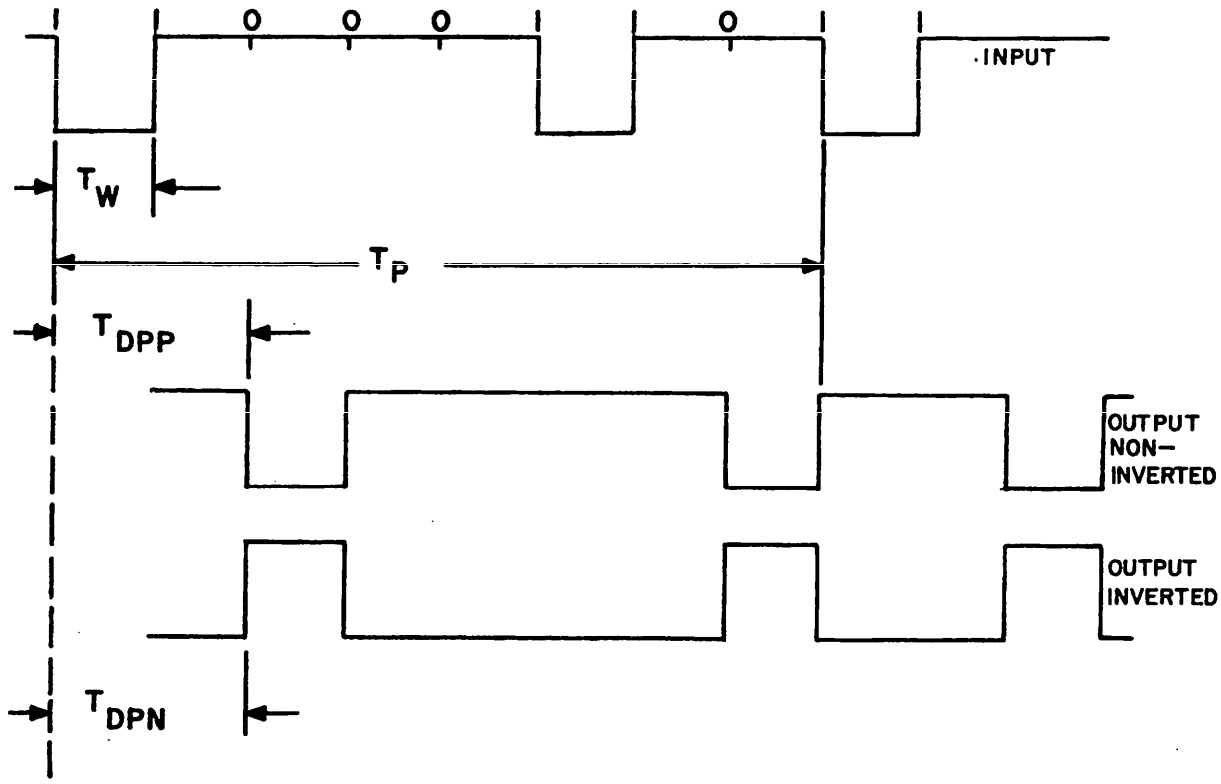


FIGURE 4. PERFORMANCE TEST SET UP

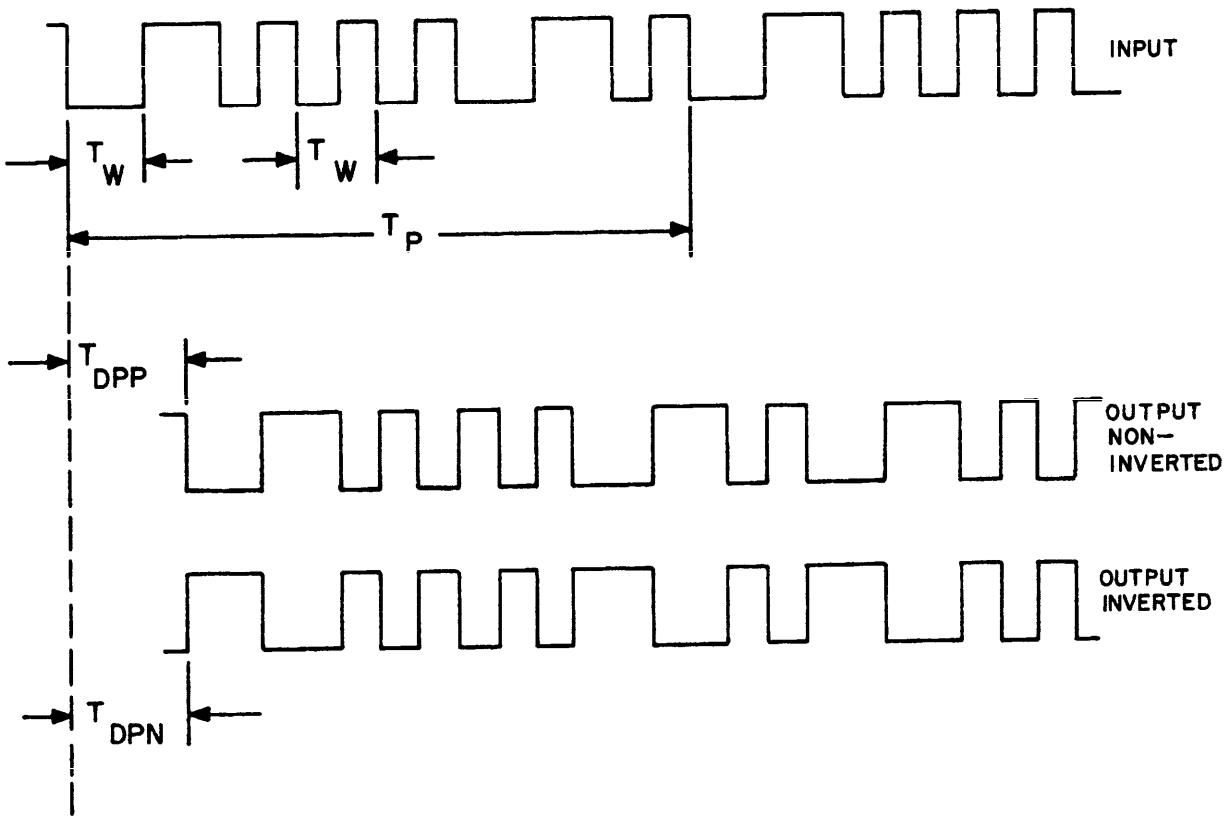
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	BIT RATE		UNIT
	16.0	18.75	kbps
T_W	62.5	53.3	MICROSECOND
T_P	500.0	426.7	MICROSECOND
T_{DPP} T_{DPN}	110.0	110.0	MICROSECOND MAXIMUM

FIGURE 5. CONDITION BASEBAND

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	BIT RATE		UNIT
	16.0	18.75	kbps
T_W	62.5	53.3	MICROSECOND
T_P	500.0	426.7	MICROSECOND
T_{DPP} T_{DPN}	110.0	110.0	MICROSECOND MAXIMUM

FIGURE 6. CONDITIONED DIPHASE WAVEFORM

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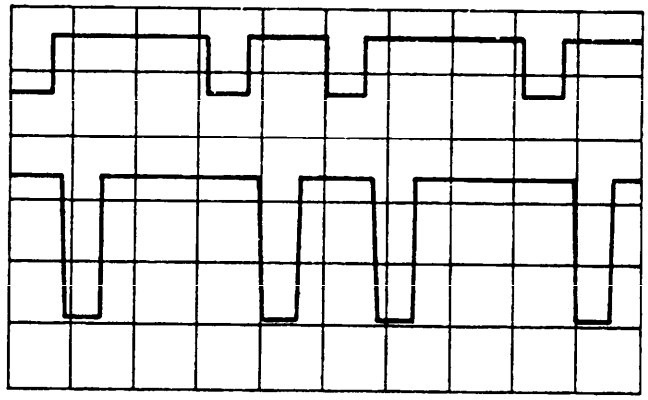


FIGURE 7. SIGNALS SHOWN AT 16.0 kbps

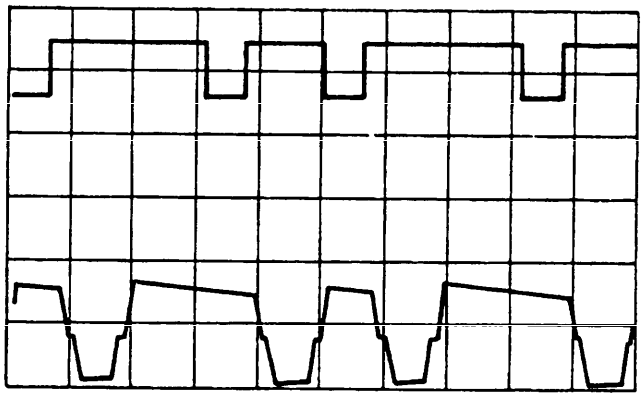


FIGURE 8. SIGNALS SHOWN AT 16.0 kbps

* AN INVERTED OUTPUT SIGNAL IS EQUALLY VALID

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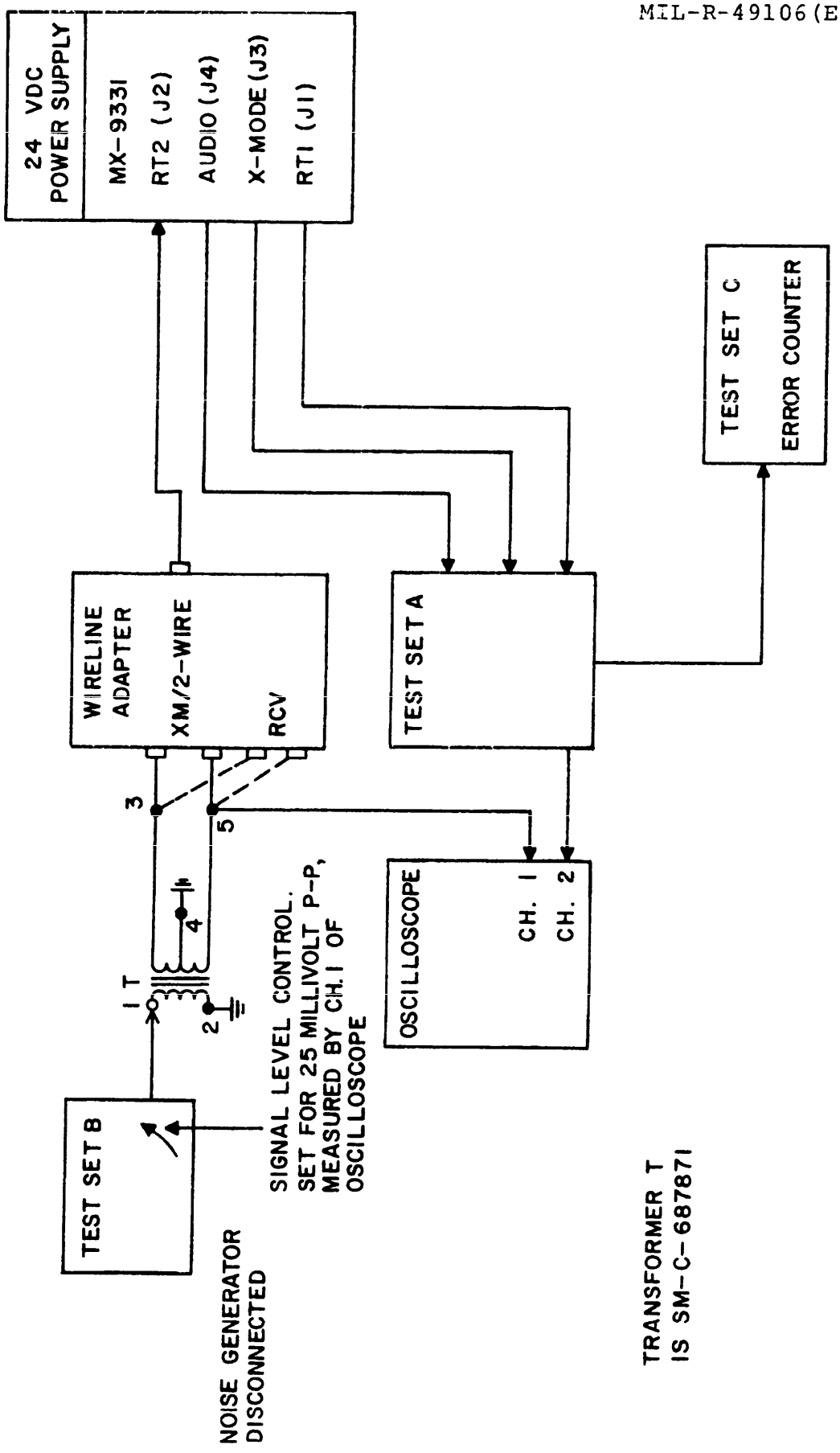


FIGURE 9. SINGLE WIRE LINE ADAPTER WITH MX-9331()/URC

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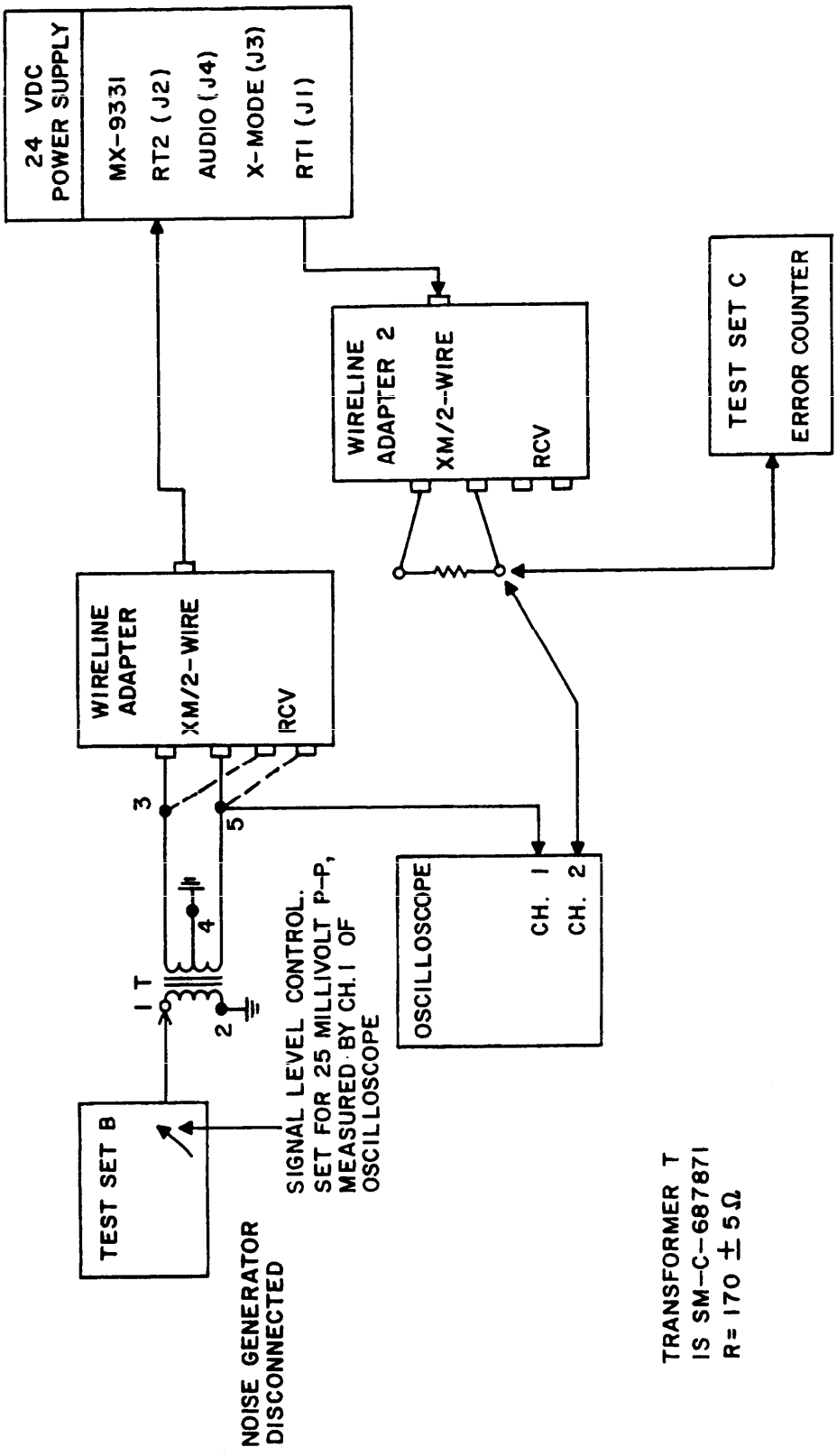


FIGURE 10. TWO WIRE LINE ADAPTERS WITH MX-9331( )/URC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-49106(EL) REPEATER, REGENERATIVE MX-9331( )/URC And AUXILIARY EQUIPMENT

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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